

CT images demonstrating the natural progression of locally advanced partially treated rectal cancer

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DESCRIPTION

A woman in her 60s presented to an outside facility for lower back pain. CT revealed a rectal mass invading her sacrum and lower pelvis alongside a surrounding abscess. Endoscopic biopsy confirmed adenocarcinoma, with no distant metastases on CT, consistent with Stage III rectal cancer. She initially underwent 5 weeks of long course neoadjuvant chemoradiotherapy (LCRT) in hopes of reducing her tumour burden before potential surgical resection. Her regimen consisted of leucovorin, fluorouracil (5-FU) and oxaliplatin (FOLFOX) and was associated with significant fatigue, neuropathy and weakness. She completed radiation therapy into the pelvis with 50.4 greys of 5-FU given in 28 fractions. On completion of LCRT, two major facilities could not guarantee cure with abdominoperineal resection due to lack of downstaging of the locally aggressive tumour. The patient was fully aware that she would eventually need a permanent colostomy regardless of which treatment option she pursued and opted for palliative care due to fear of living with a permanent colostomy much earlier in her residual lifespan and fear of possibly disfiguring pelvic surgery. She declined adjuvant chemotherapy due to the strong side effects she suffered from her initial course of chemotherapy. Since her initial diagnosis she has developed numerous presacral abscesses secondary to local neoplastic tissue invasion requiring CT-guided drainage (figure 1). One year after initiating palliative care, she underwent permanent diverting colostomy due to large bowel obstruction.

She was referred to our office due to worsening lower pelvic and sacral pain. A CT with rectal contrast revealed complete disappearance of the posterior rectal wall, mesorectum and one large cavity in the low pelvis consisting of tumour

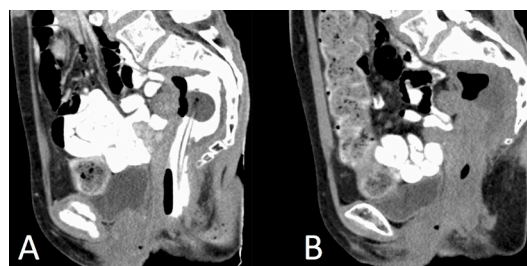


Figure 2 Sagittal CT images with (A) and without (B) rectal contrast revealed complete disappearance of the posterior rectal wall and one large cavity in the low pelvis consisting of tumour infiltration into the presacral tissues.

infiltration onto the presacral periosteum. In some areas, the rectal contrast superimposed the sacral bone consistent with localised tumour infiltration of the sacral nerve plexus, likely the source of her pain (figure 2).

Only the anterior upper third of the rectum is covered in a protective serosal layer. The thin-layered fascia propria is the sole structure separating the majority of the rectum from the surrounding lymphatic-rich mesorectum.¹ Given the lack of protective layers to insulate a growing rectal tumour from the mesorectum, poor oncological outcomes are associated with tumour mass extending up to 1 mm from the fascia propria.² LCRT followed by en bloc surgical resection is the standard therapy for locally advanced rectal cancers (stage IIB–IIIC).^{3,4} Although our patient underwent LCRT, her primary tumour did not downstage and continued to invade into the mesorectum. Thus, she was not a candidate for mesorectal excision and her disease progression mirrored that of untreated rectal carcinoma. We were able to capture the rare outcome of complete erosion into the mesial rectum with the disappearance of the posterior rectal wall and mesentery on CT, as this patient's tumour was unresponsive to

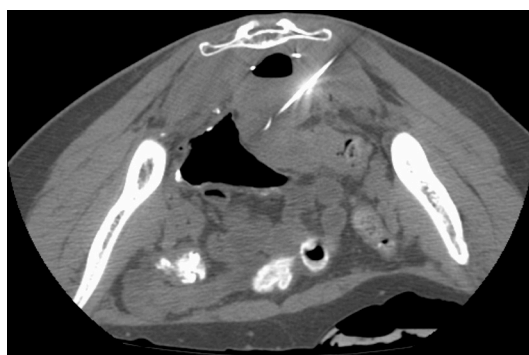


Figure 1 CT-guided drainage of presacral abscess secondary to rectal carcinoma.

Learning points

- ▶ The definitive treatment for non-metastatic rectal cancer is surgical excision.
- ▶ If surgical treatment is refused and long course neoadjuvant chemoradiotherapy does not successfully downstage the primary tumour, locally advanced rectal cancer can lead to complete erosion into the mesial rectum with the disappearance of the posterior rectal wall and mesorectum.



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LCRT and she initially declined surgical treatment. Eventually, rectal carcinomas will penetrate local lymphatics and vasculature, resulting in metastasis to distant organs. This is typically fatal. Our patient would ultimately pass from complications due to bone metastases and malnutrition on home hospice.

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Case reports provide a valuable learning resource for the scientific community and can indicate areas of interest for future research. They should not be used in isolation to guide treatment choices or public health policy.

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